Impact of Water Availability on Tomato Plant Growth and Development

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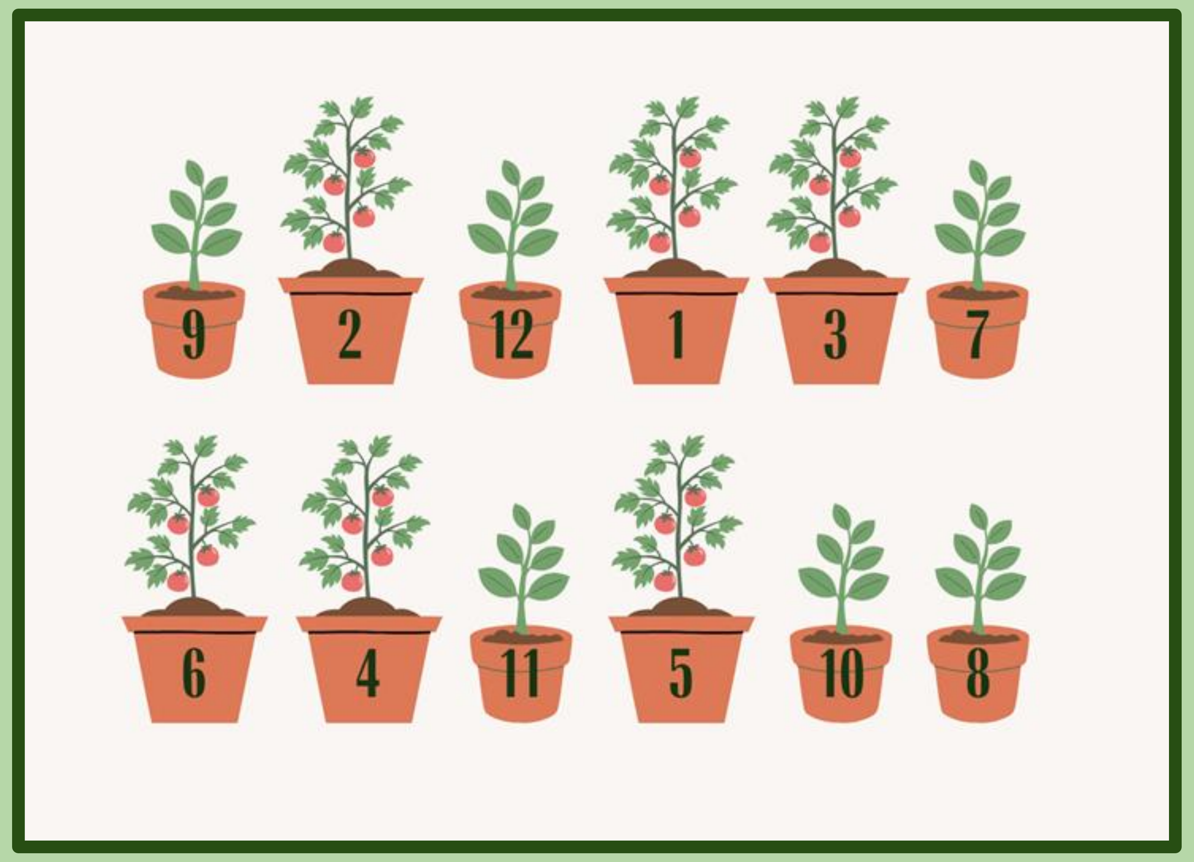
Introduction

California is expected to experience increasing drought severity and frequency as a result of climate change. This, along with state water policies, will continue to limit water availability for agricultural production statewide. In 2023, tomato production contributed $2.01 billion to California's agricultural economy[1]. This study dives into how water availability affects the growth and yield of processing tomato plants[2]. To simulate this, we grew tomatoes in different sized nursery pots in a greenhouse. Our team was hoping that by researching resilient tomato production strategies, we can provide more information on the effect of water availability when growing tomatoes to help California's agricultural systems that rely heavily on this crop.

Hypothesis

If a processing tomato has access to a higher volume of water, then it will have higher fruit yield production, more flowers, and increased height and biomass.



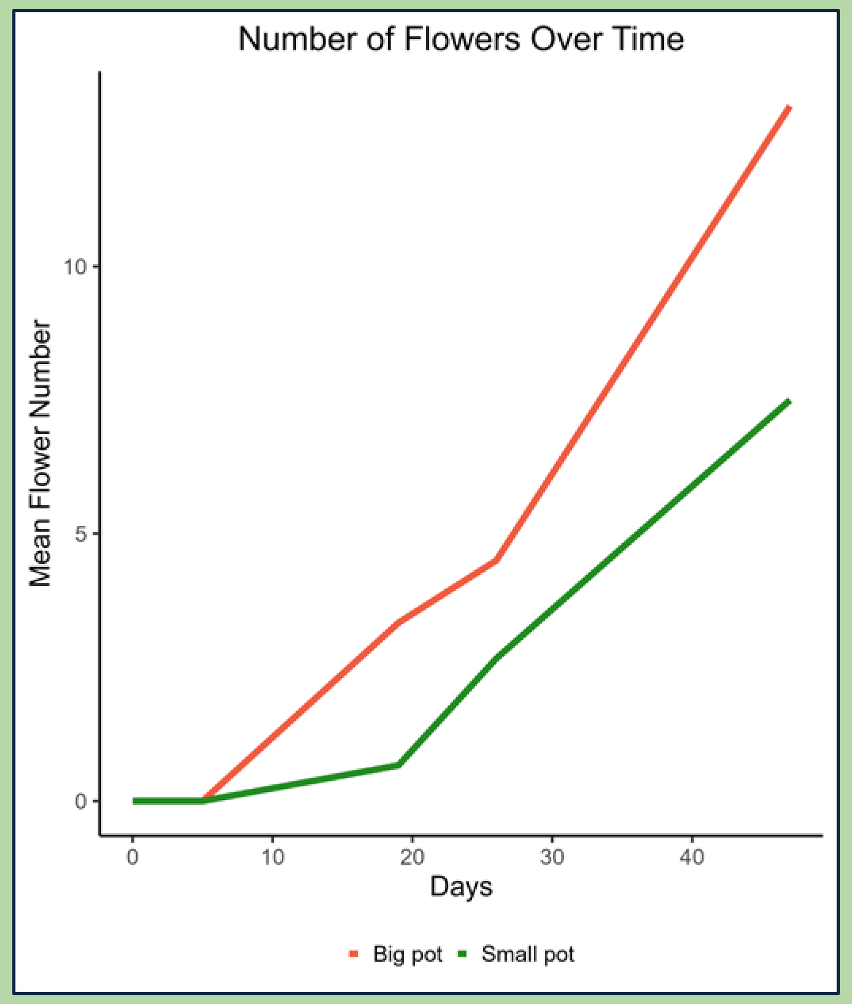
Materials

- 6 one-gallon "big" pots versus 6 half-gallon "little" pots
- Randomly placed on a bench inside a greenhouse
- Experiment ran for 6 weeks with data collection occurring approximately every other week
- Collected data for the plant's height, leaf number, flower number, fruit number, number of inflorescence, and the number of flower buds
- All pots were irrigated with the same frequency and volume
- The pot sizes simulated different irrigation volumes
- Completed our data analysis and visualization using R

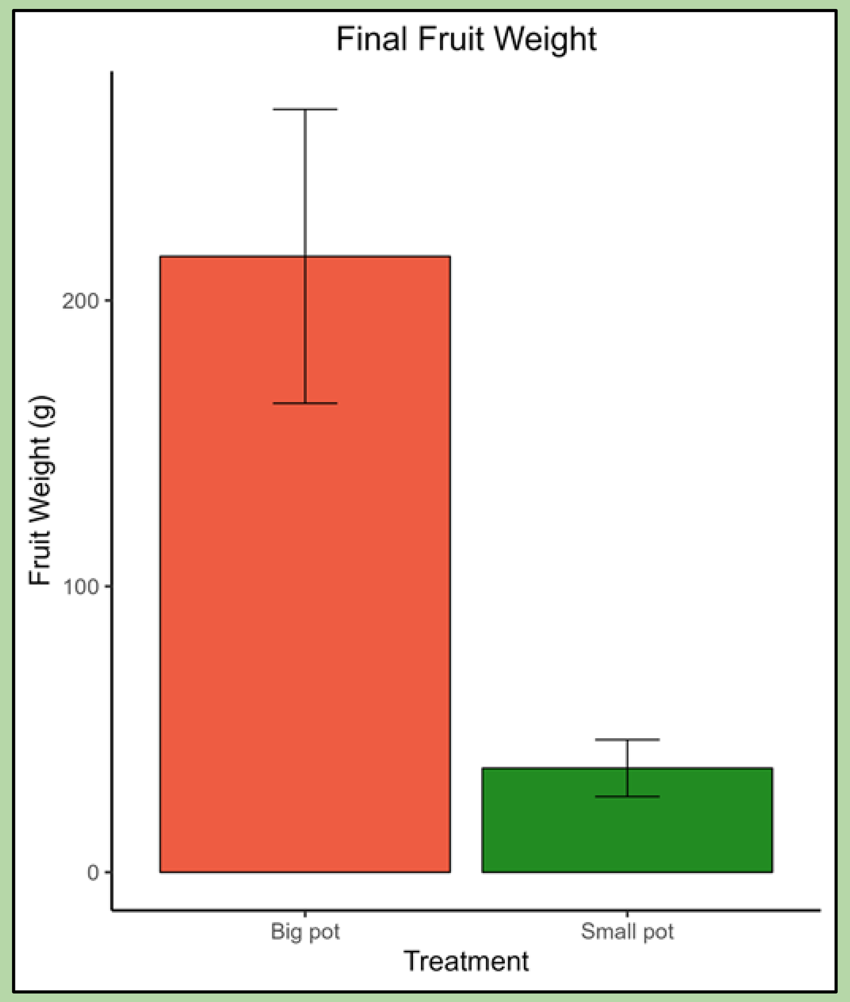
Methods

- Both treatments were the same black nursery pots; the only difference was their size
- Grew processing tomato plants that started from transplants, not seeds

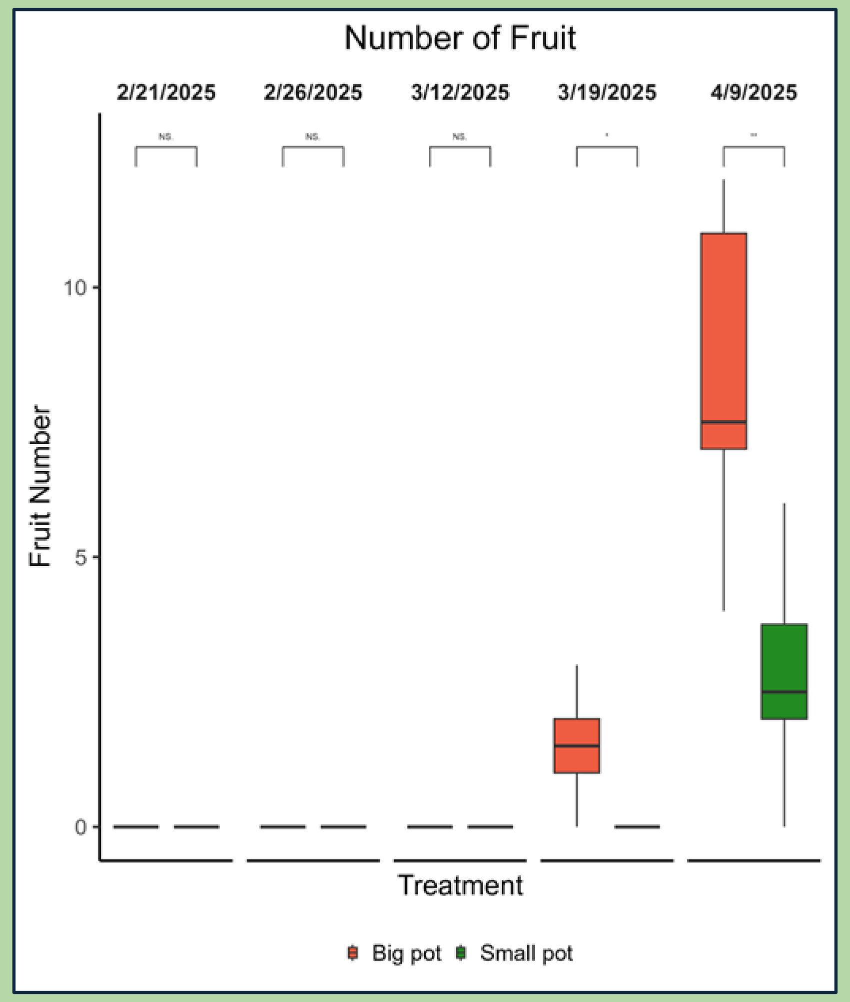
RESULTS



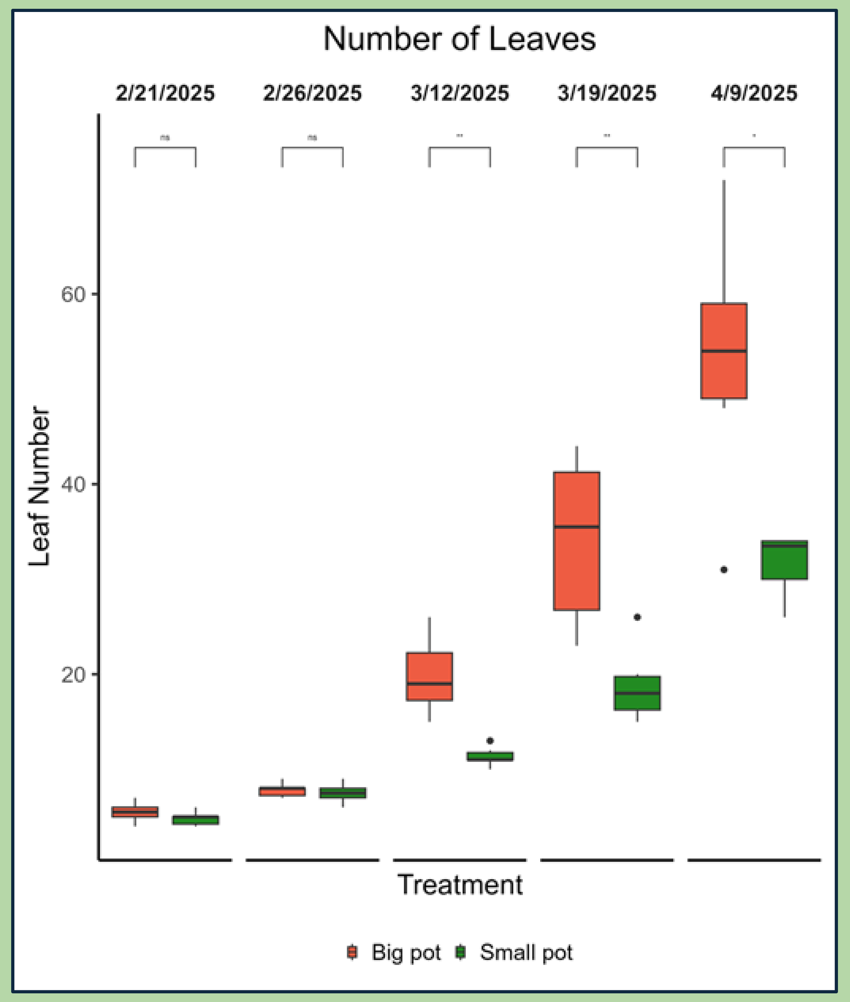
The graph represents the mean flower number throughout each data collection period, showing that the big pots overall produced more flowers.



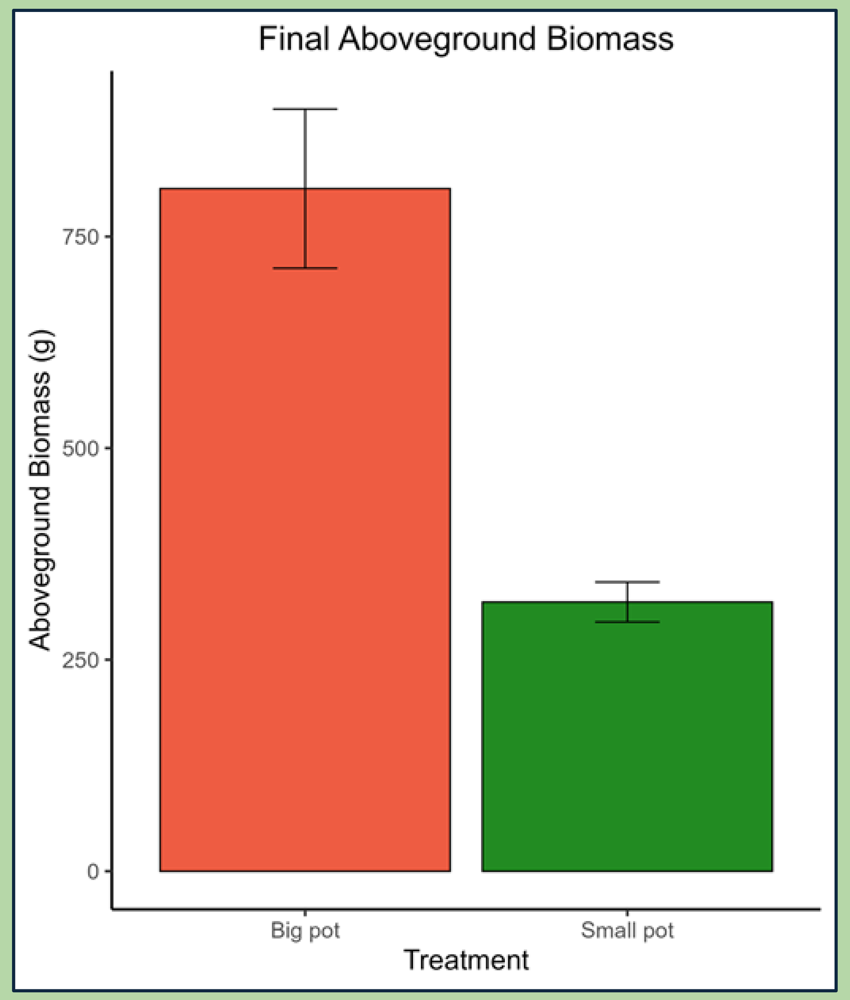
This is our final yield data taken on day 53. There are error bars at the top of each bar with the big pot's data in red while the small pot's data is in green. The tomatoes were ripe and ready for harvest, and this graph gave us an idea of what the final yield would look like.



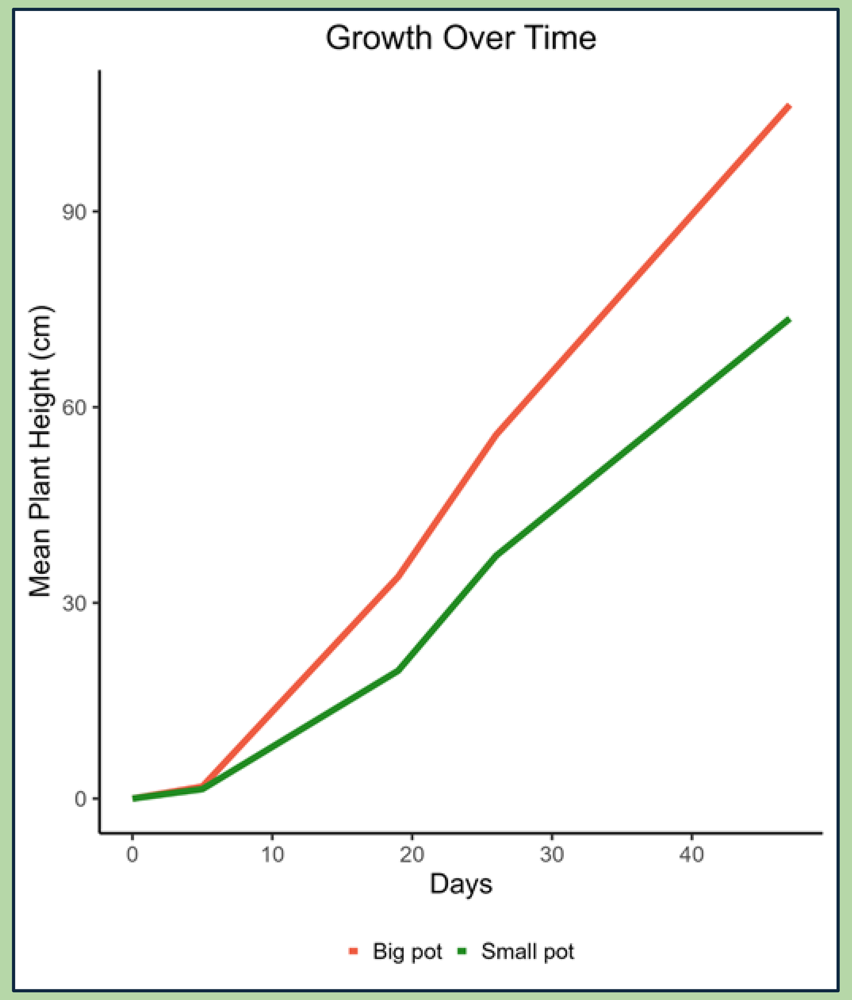
The graph displays the number of fruit grown over time on the tomatoes in the big pots vs. the small pots.



Our data showed that the tomato plants grown in bigger pots grew more leaves and showed increased development.



The biomass was taken at the very end of our experiment while including error bars. The aboveground biomass is the total biomass between the vegetative and fruit biomass.



This graph illustrates the tomatoes' growth over time. It indicates how the plants grown in the bigger pots developed more rapidly.

Discussion

Several factors likely influenced our results and why the tomatoes grow better in bigger pots.

Improved Water Retention:

- Bigger pots hold more soil, which increases water-holding capacity.
- It helps maintain consistent moisture levels, supporting continuous growth and fruit development—critical for tomatoes.

Better Nutrient Availability:

- In small pots, nutrients are quickly depleted or may become concentrated and imbalanced, which could end up stressing the plant.

Understanding how pot size influences tomato growth helps researchers model how soil volume and water availability affect field-grown tomatoes, specifically under drought conditions3. This supports the development of efficient growing practices that optimize yield while conserving water.

It is unclear here whether our results are due to water limitations alone or whether other factors such as root growth may have contributed4. Future research should also consider how these treatments affect root growth. I would have been curious to see if our bigger pots have improved root development.

Both photos display data collection by counting the plant's leaves, flowers, flower buds, number of inflorescence, and measuring its overall height from the top soil to the meristem.

References

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